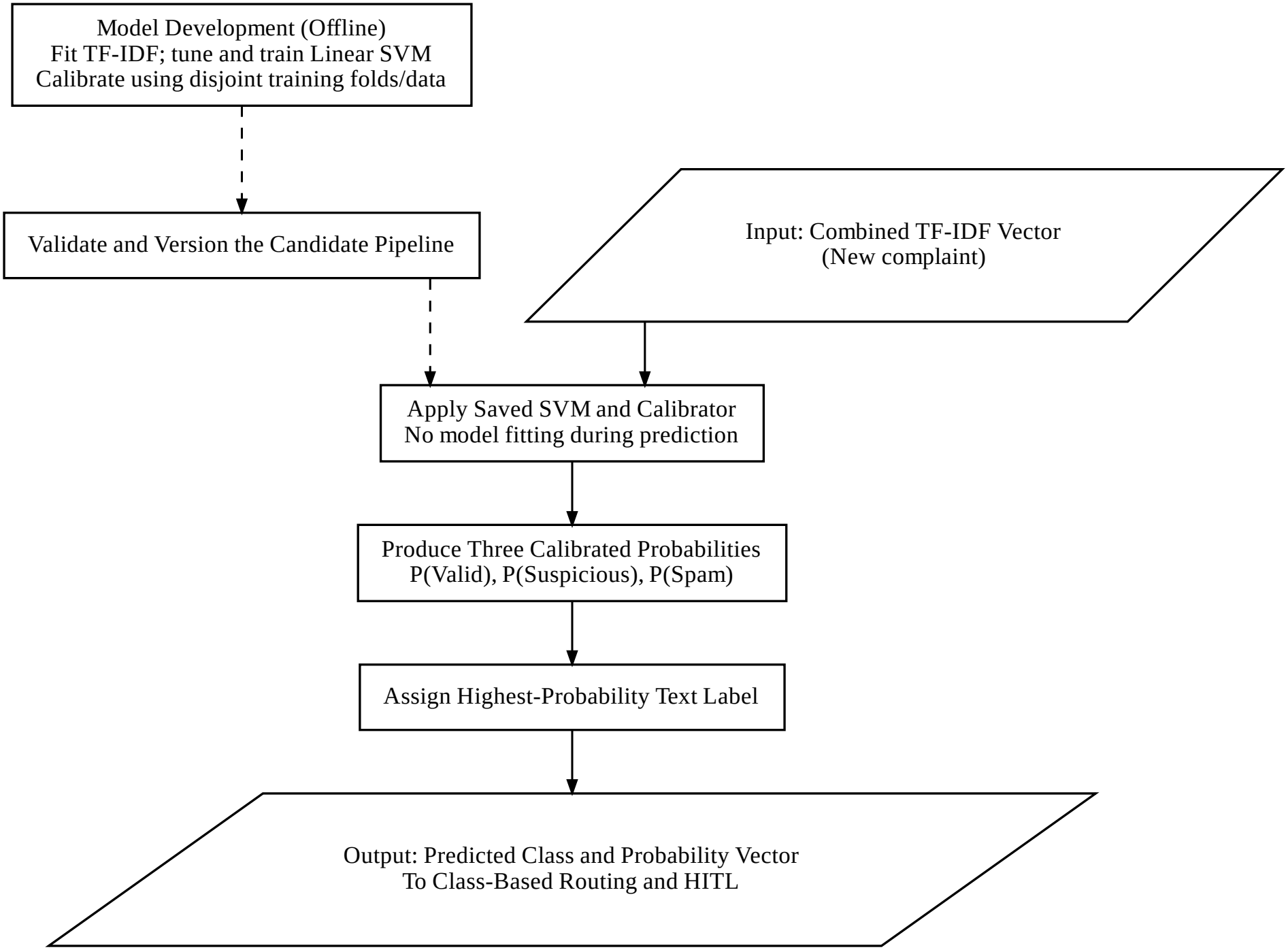
Model Development (Offline)
Fit TF-IDF; tune and train Linear SVM
Calibrate using disjoint training folds/data
Validate and Version the Candidate Pipeline
Input: Combined TF-IDF Vector
(New complaint)
Apply Saved SVM and Calibrator
No model fitting during prediction
Produce Three Calibrated Probabilities
P(Valid), P(Suspicious), P(Spam)
Assign Highest-Probability Text Label
Output: Predicted Class and Probability Vector
To Class-Based Routing and HITL